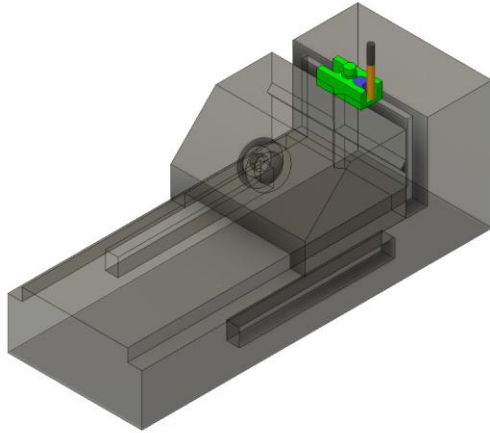


Desktop CNC Milling Machine: Cutting Crank-Slider SLIDER

I: General Guidelines



GENERAL SAFETY AND OPERATIONAL INSTRUCTIONS – READ BEFORE USE:

- Users must demonstrate proper personal protective equipment (PPE) use before performing work on this machine.
- Users must wear appropriate PPE (safety glasses, ear protection and mask when necessary) when operating this machine.
- NEVER operate the machine tool without all safety mechanisms (enclosure, door switch, emergency stop, limit switches) connected, complete, and fully operational.
- **NEVER BYPASS ANY SAFETY EQUIPMENT.**
- NEVER place hands, head, or other body parts inside the machine tool enclosure while spindle is powered and door switch / emergency stop are disengaged.
- NEVER handle sharp tools in the spindle without the emergency stop engaged.
- During ANY milling operation, if the cutting tool shows excessive wear or excessive chip build up, or if the spindle's RPM changes noticeably (this is often audible), quickly feed hold or emergency stop the machine (depending on severity) to inspect and determine if it is safe to continue.
- Always allow the spindle to reach its target speed before entering a cut.
- Always double-check setup steps and part programming to minimize machine accidents.
- Always watch and listen to EVERY milling operation closely. Keep a hand resting over the emergency stop button: quick reactions are necessary in mitigating damage to the machine and tools.

II: Operator Checklist

Operator name: _____

Workpiece: _____

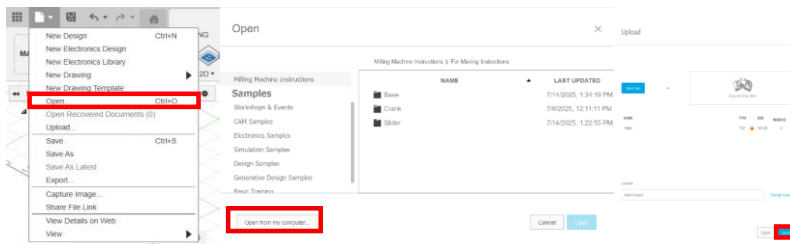
Date: _____

Before running the program		
OP10	OP20	
☐	☐	Program has been simulated in CAM
☐	☐	All tool offsets are set per the setup sheet
☐	☐	Workpiece is clamped in the proper position
☐	☐	Correct program is selected on the machine
		Work offsets are set for:
☐	☐	X axis
☐	☐	Y axis
☐	☐	Z axis

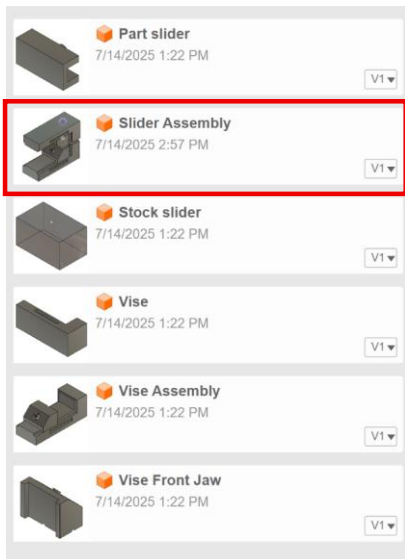
For each tool while running the program		
OP10	OP20	
☐	☐	Check position as the tool approaches the part

III. Part Programming (Autodesk Fusion)

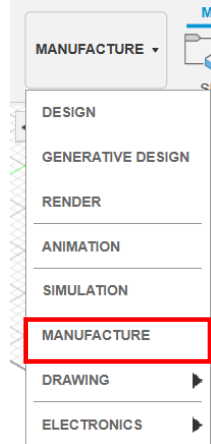
1. Start Fusion and open the provided file. Select the *open* option from the drop-down menu in the top left corner of Fusion. Select *Open from my computer ...* in the following prompt. Locate *Slider.f3z* and select *upload*.



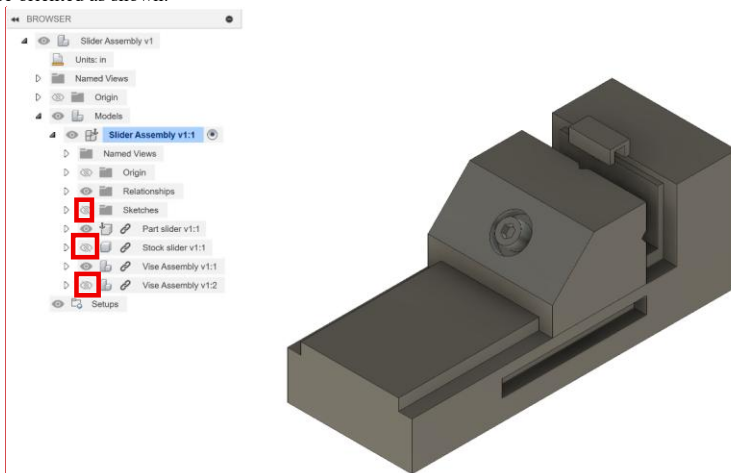
2. After uploading the file, three parts and an assembly will be in the folder on the left-hand side of the program. Open the *Slider Assembly* file.



- From the drop-down menu, select the *Manufacture* tab to begin toolpath generation.



- Hide the sketches under *Slider Assembly v1:1*. Hide the component *Stock slider v1:1*. Hide *Vise Assembly v1:2*. To hide components and sketches select the eye outlined in red. Ensure your view cube and assembly are oriented as shown.



Commented [TW1]: Need to hide vise assembly v1:2

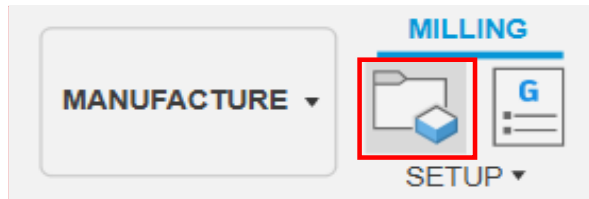
Commented [TW2]: Add initial view of assembly

Commented [TW3R2]: Fix the tree menu

Commented [TW4R2]: Hide relationships

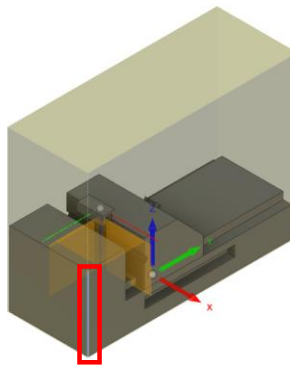
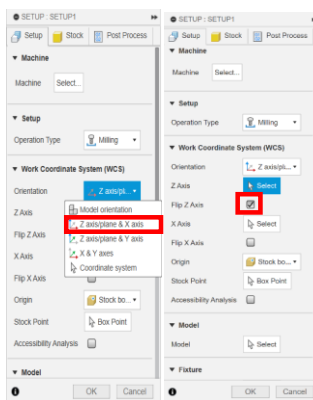
Commented [TW5R2]: Update hiding of assembly to one button

5. Create a new setup.



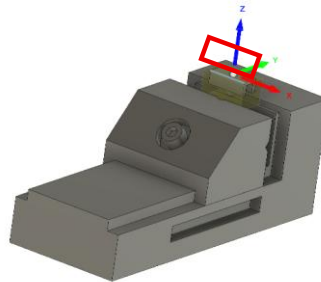
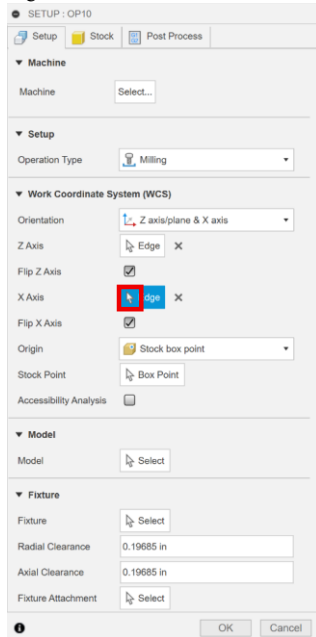
Commented [TW6]: Put "go to manufacture tab" before the hiding of the views. Decouple step 4.

6. From the Orientation drop-down menu, select *Z axis/plane* & *X axis*. Rotate the assembly and select the back edge outlined in red for the Z axis. Check the *Flip Z Axis* box.

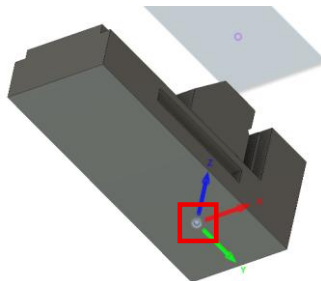
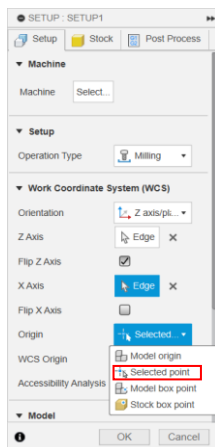


Commented [TW7]: Choose a better edge

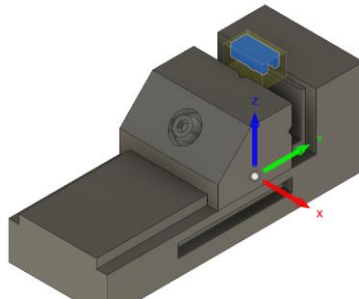
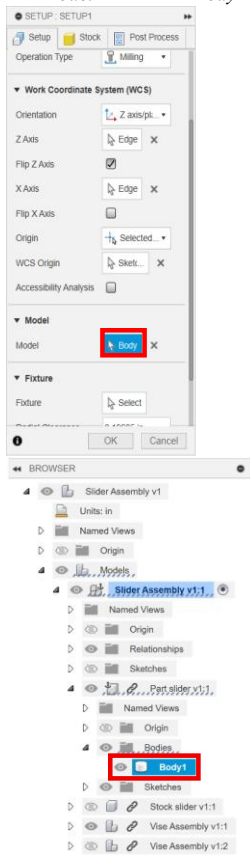
7. After selecting the *Z* axis, select the *X* axis as the horizontal edge outlined in red. Check the *Flip X Axis* box.



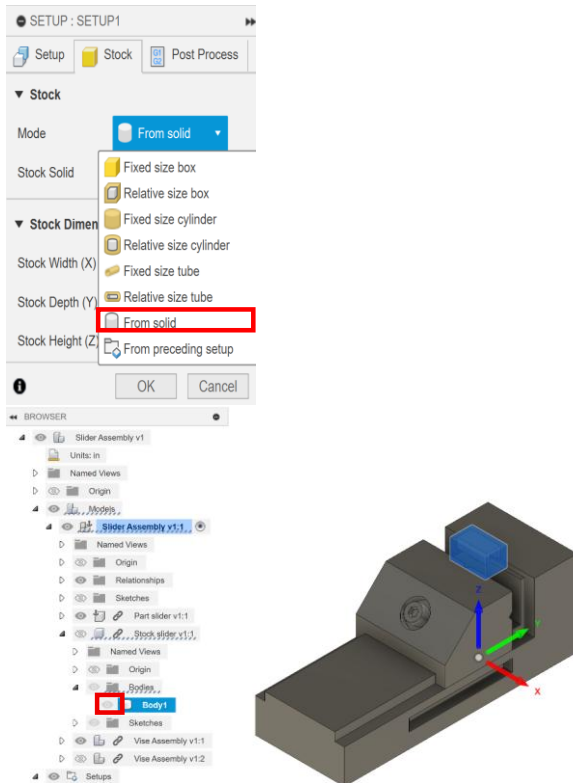
8. Select “Selected point” under the *Origin* drop-down menu. Select the point on the bottom of the vise outlined in red. You may need to make sketches visible.



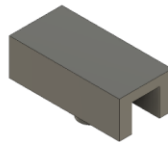
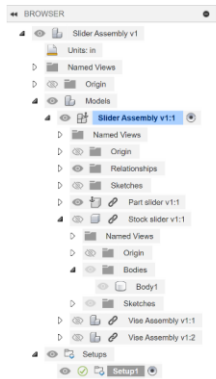
9. Select *Model* and choose *Body1* under *Part slider v1:1*.



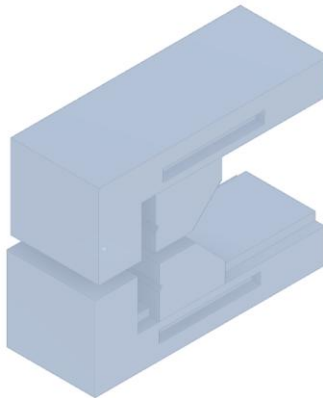
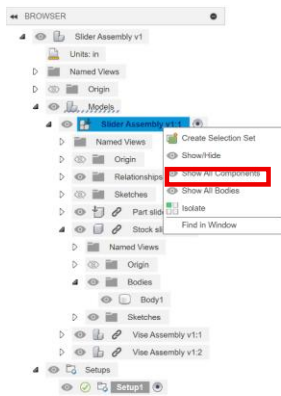
10. Under the *Stock* tab, select *From solid* for the stock mode. Chose *Body1* under *Stock slider v1:1* . Then click *OK*.

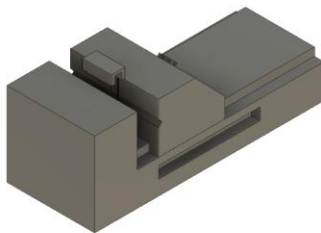
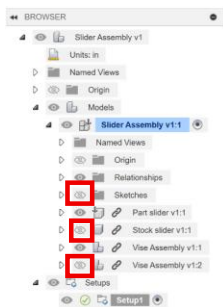


11. Once you have clicked *OK* you may have noticed that your part slider is the only body that appears in your software. Right click on *Slider Assembly v1:1* and select *Show All Components*. Once that is done, repeat Step 3.



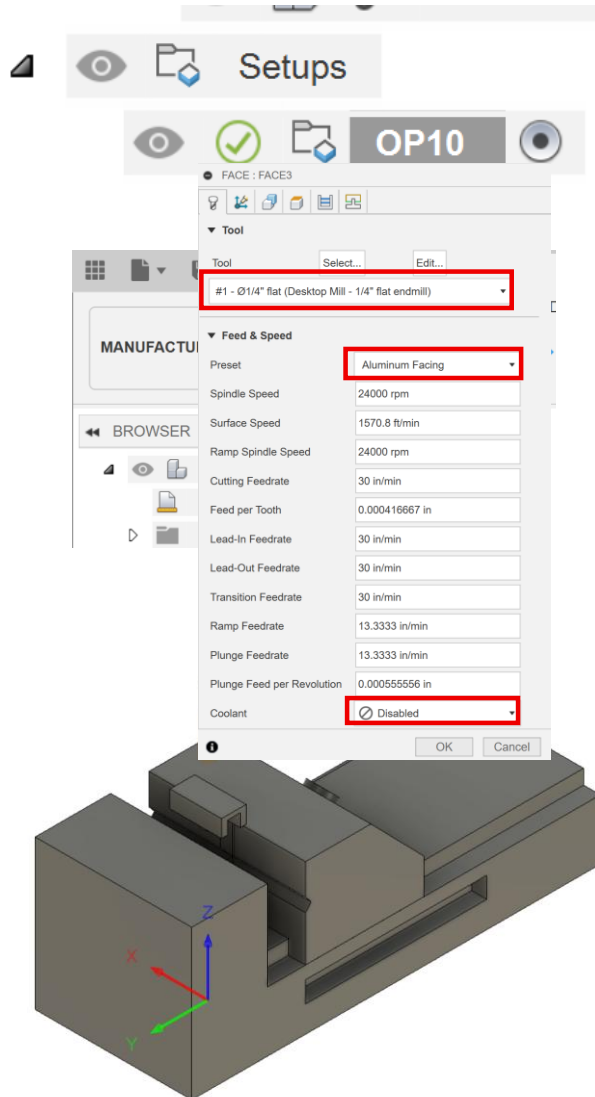
Commented [TW8]: Make figures larger so we can read



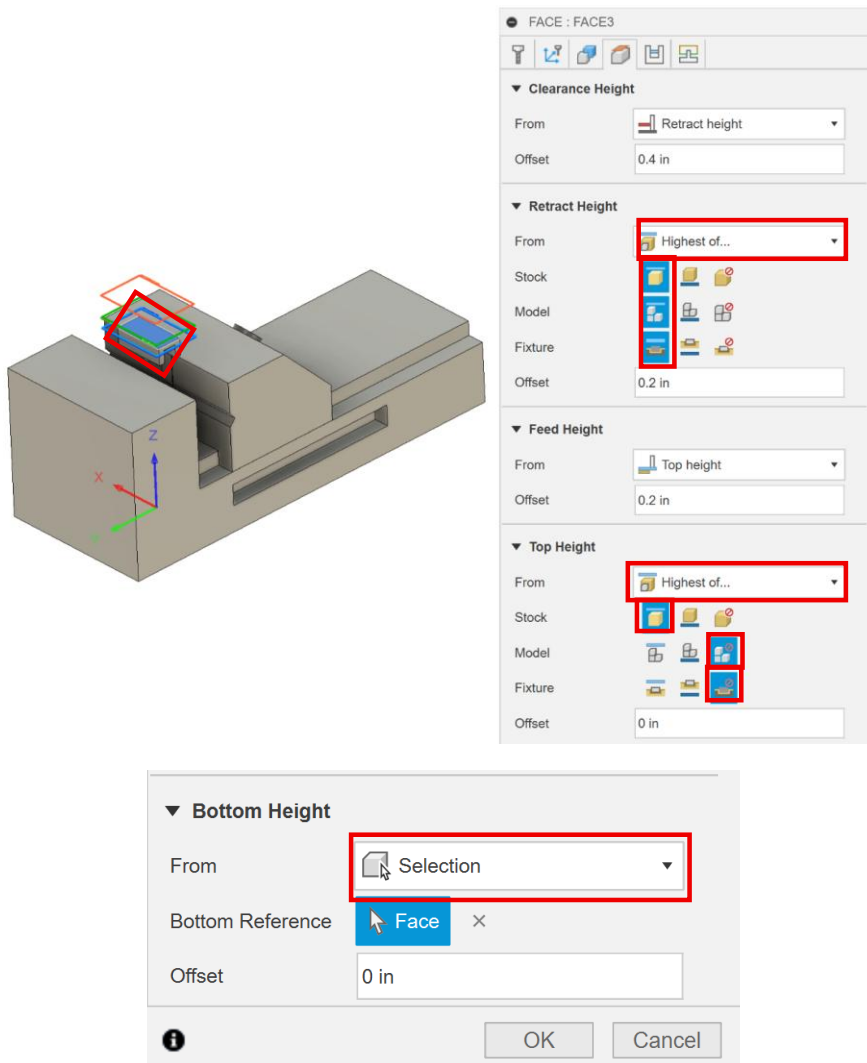


12. Slowly double click *Setup1* and rename to *OP10*. Next select *2D Face* and select the 0.25-inch diameter tool within the library. Make sure *Aluminum facing* preset is selected.

Commented [TW9]: Should say "Aluminum facing"



13. In Retract Height, change from clearance height to *Highest of...* and select the top option. In *Top Height*, select from *highest of* and choose *top for stock* and ignore for model and fixture. In *Bottom Height*, choose *selection* and pick the top face of the part.



14. Finally, select *climb* for direction in the *Passes* tab and click *ok*.

Passes

Tolerance: 0.0004 in

Pass Direction Reference: [Select](#)

Pass Direction: 0 deg

Pass Extension: 0.1254 in

Stock Offset: 0 in

Stepover: 0.175 in

Direction: **Climb**

Order For Shorter Links: ☐

From Other Side: ☐

Use Chip Thinning: ☐

☐ Multiple Depths

☐ Stock to Leave

OK Cancel

15. Under 2D, select 2D-Adaptive clearing, and make sure the 1/4" is selected. Change the preset to Aluminum Contour Roughing. In geometry tab, select the top of the slider for pocket selections.

2D ADAPTIVE : 2D ADAPTIVE3

Tool

Tool: [Select...](#) [Edit...](#)

#1 - Ø1/4" flat (Desktop Mill - 1/4" flat endmill)

Feed & Speed

Preset: **Aluminum Contour Roughing**

Spindle Speed: 24000 rpm

Surface Speed: 1570.8 ft/min

Ramp Spindle Speed: 24000 rpm

Cutting Feedrate: 48 in/min

Feed per Tooth: 0.00066667 in

Lead-In Feedrate: 48 in/min

Lead-Out Feedrate: 48 in/min

Transition Feedrate: 48 in/min

Ramp Feedrate: 13.3333 in/min

Plunge Feedrate: 13.3333 in/min

Plunge Feed per Revolution: 0.00055556 in

Coolant: Disabled

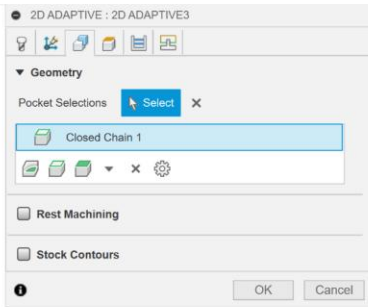
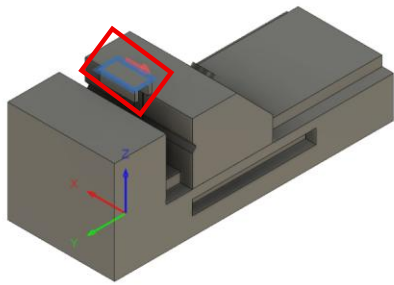
OK Cancel

TURNING **ADDITIVE**

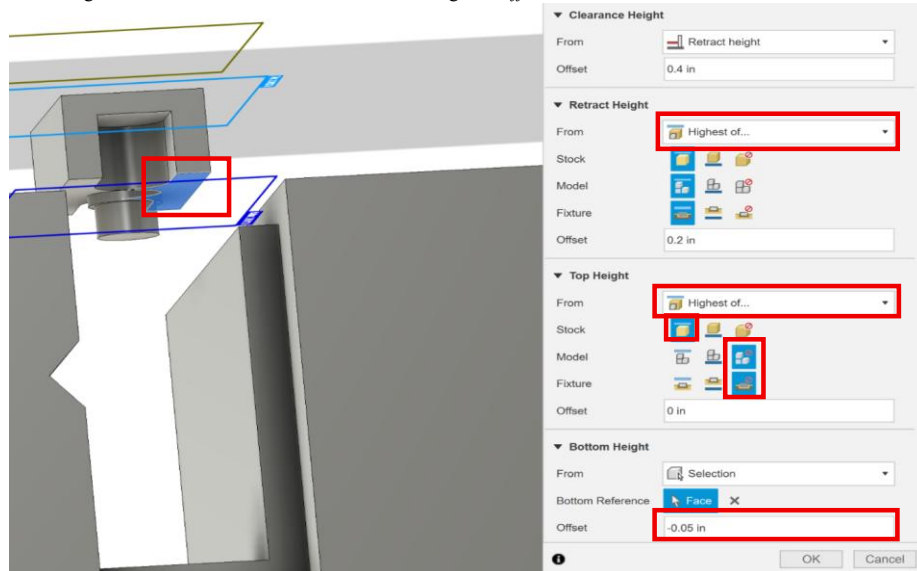
2D ▾

2D Adaptive Clearing

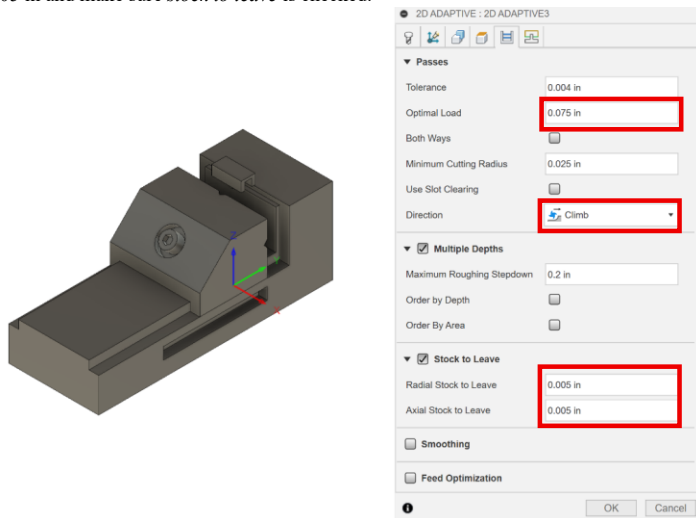
2D Pocket



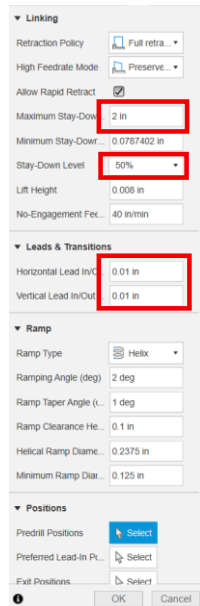
16. Switch *Retract height* and *Top height* to *Highest of*. Change the *Bottom height* to selection and select the bottom right face, zoom in as needed to select it. Change the *offset* to -0.05 in.



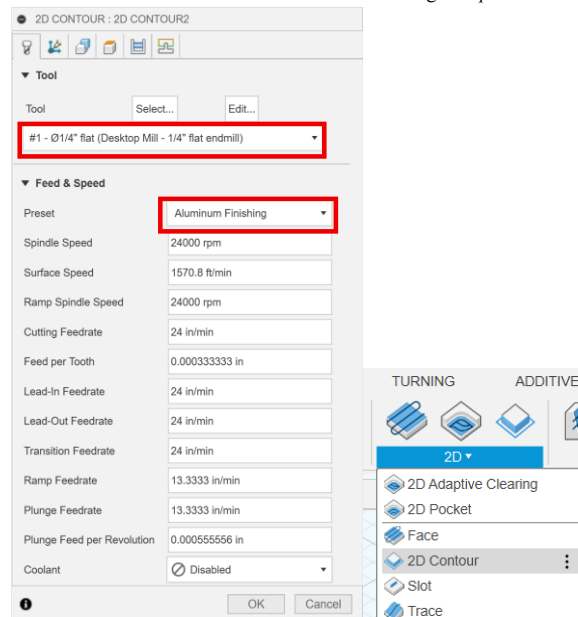
17. Under the *Passes* tab, change the *optimal load* to 0.012 in, *radial stock to leave* and *axial stock to leave* to 0.005 in and make sure *stock to leave* is checked.



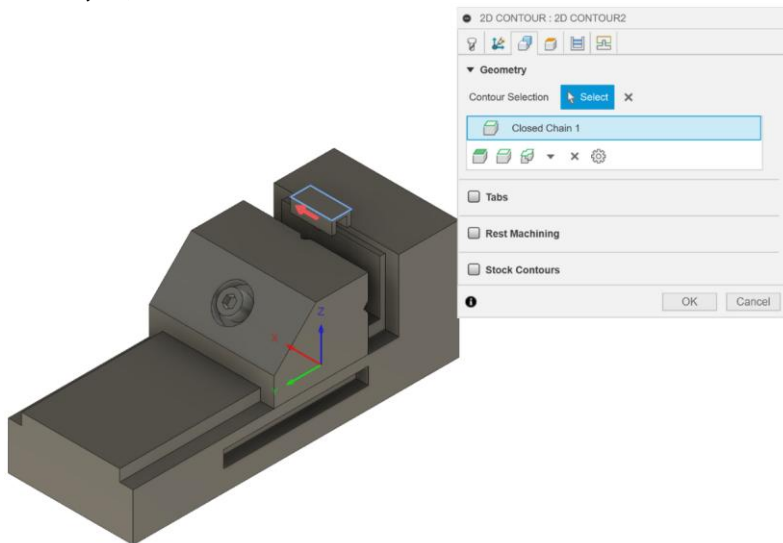
18. In the *linking* tab, change the *maximum stay-down* to 2 in, and choose 50% *stay-down level*. Then, change the *leads and transitions* to 0.01 in. Then click *ok*.



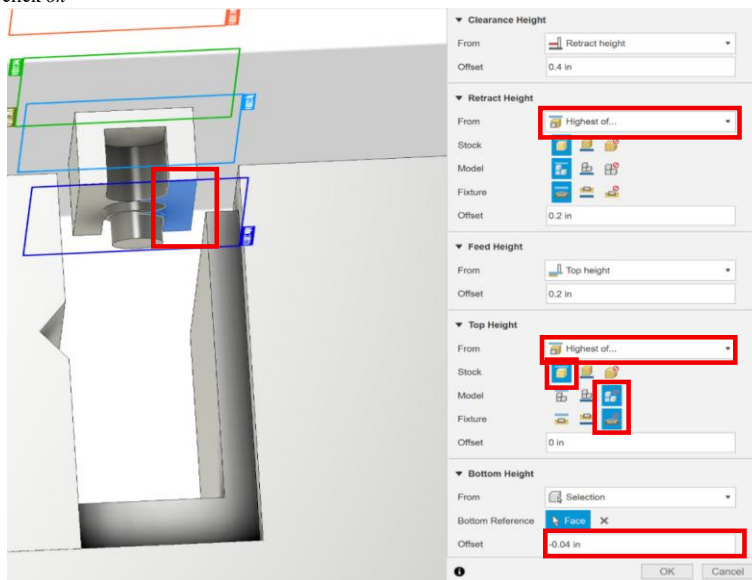
19. Next, select *2D- Contour*. Make sure the same 1/4" tool is selected. Change the *preset* to *Aluminum Finishing*.



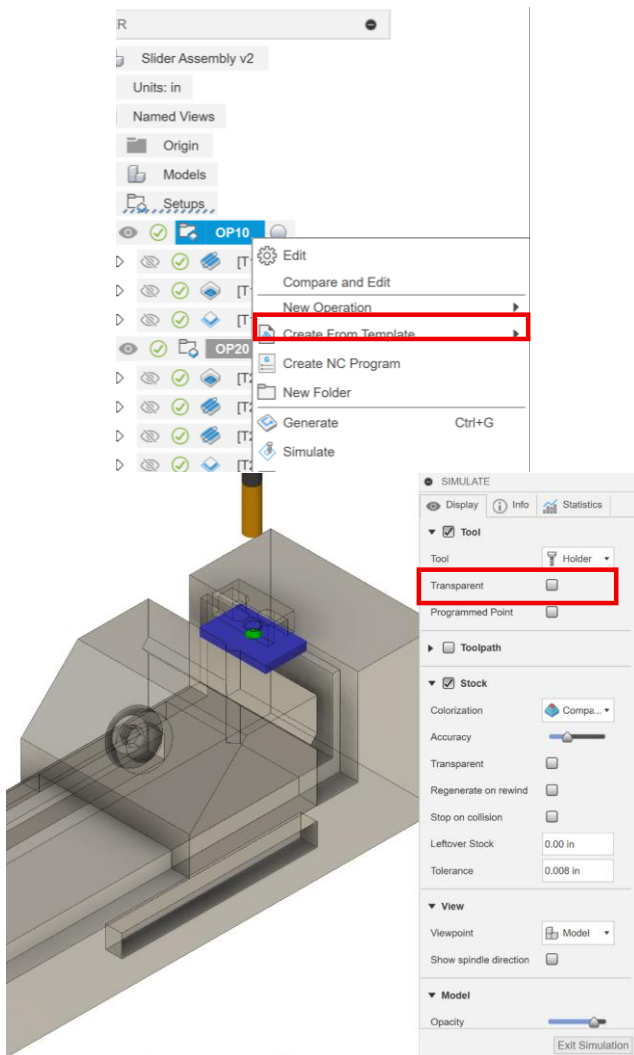
20. Under *Geometry* tab, select the contour of the slider's base.



21. Under *heights*, change the *retract height* and *top height* to the *highest of*. Then switch to *selection* in *bottom height* and select the face on the top right of the slider. Zoom in if needed. Also, set an *offset* of -0.04 in. Then click *ok*



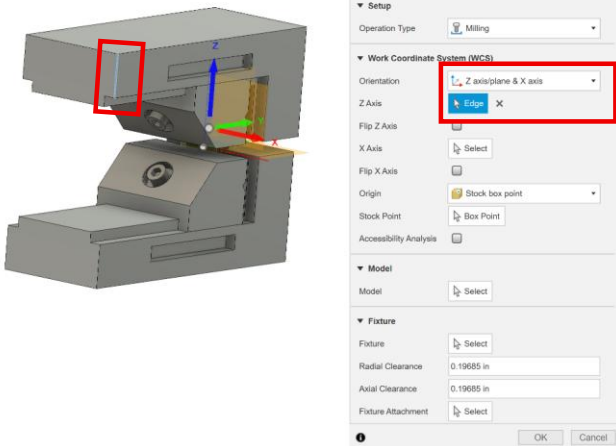
22. *OP10* should be finished at this point. Right click on *OP10* and select *Simulate*. Under *Colorization* choose *Comparison*. Press the play button at the bottom center of the window. Adjust the *Accuracy* slider until simulation runs smoothly. Once the simulation is complete, the part should be green if programmed correctly. Note: If *Accuracy* is set to the lowest value, blue may appear around the part.



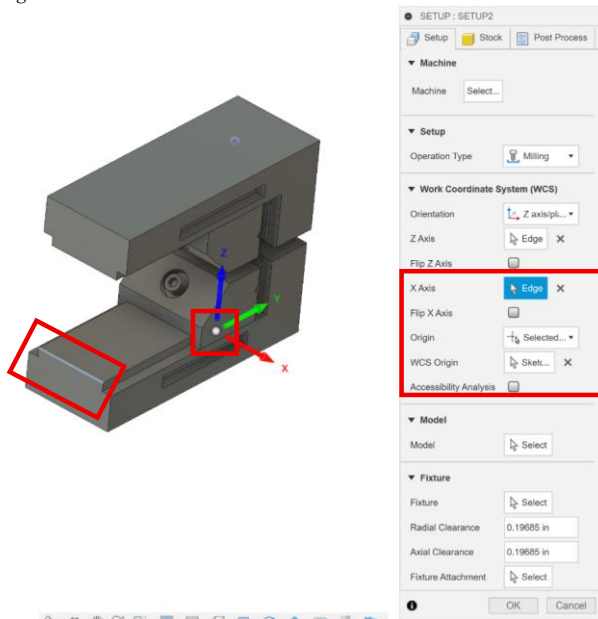
23. After finishing OP10, click on the *view* button of the *vise assembly v1:2*, and click on *setup* and *new setup* to make the CAM for OP20. Ensure your view cube and assembly are oriented as shown.



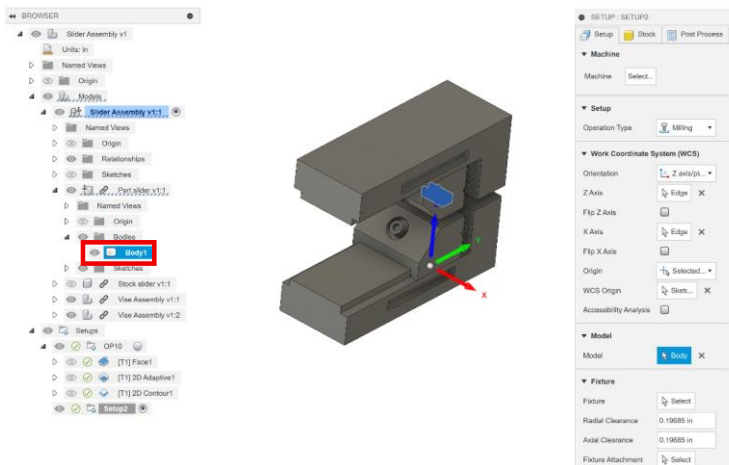
24. Change the orientation to *Z axis/plane and X axis* and select the edge highlighted in blue on the vise.



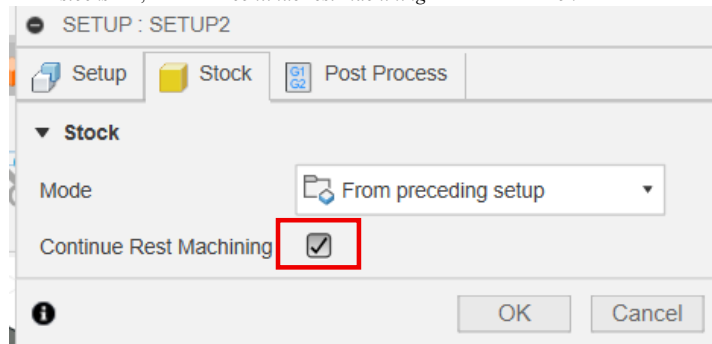
25. For the *x* axis, select edge highlighted in blue on the vise on the bottom. Then change the origin to the *selected point* option and click on the blue dot on the base. You should turn *sketches* on to select the point for the *origin*.



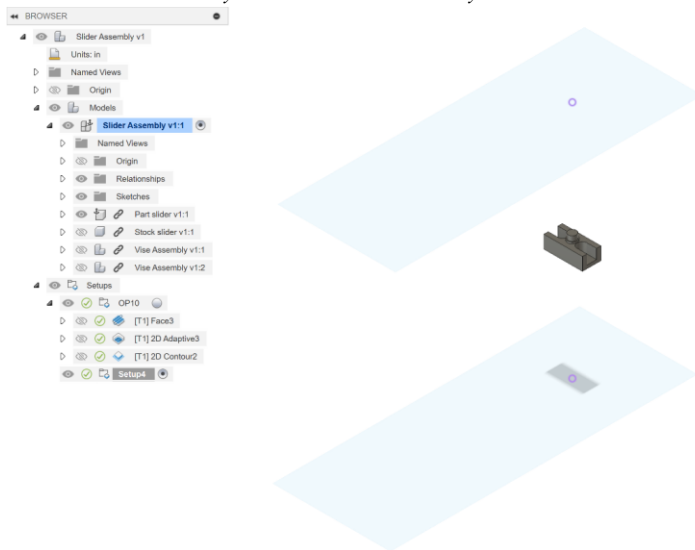
26. For the model, choose *Body 1* under *Part slider v1:1, bodies*.

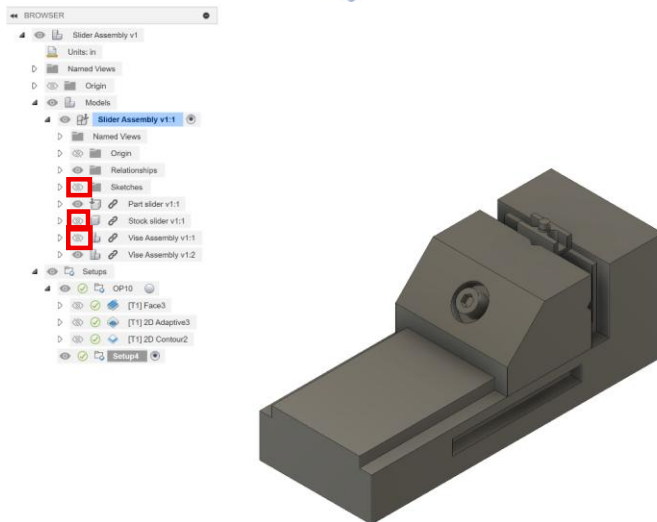
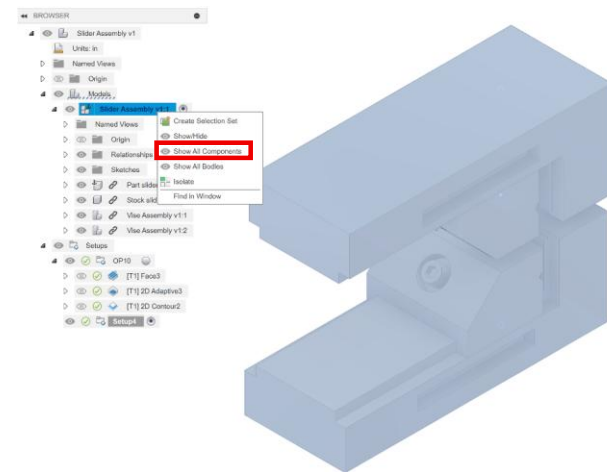


27. In the *stocks* tab, check the *continue rest machining* tab. Then click *ok*.



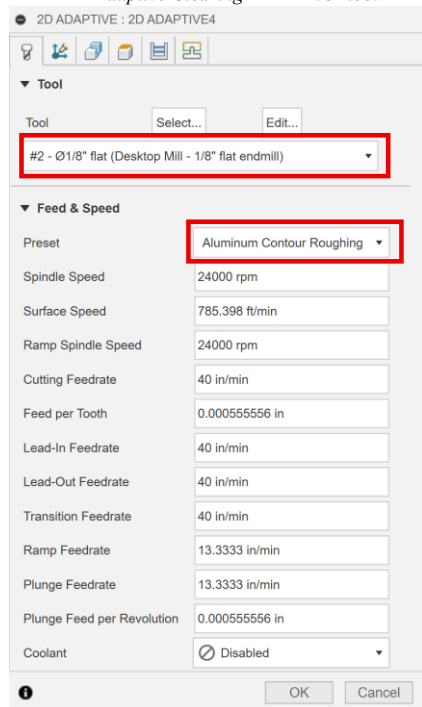
28. Again, you may have noticed that your part slider is the only body that appears in your software. Repeat step 10. But hide *Vise Assembly v1:1* instead of *Vise Assembly v1:2*.



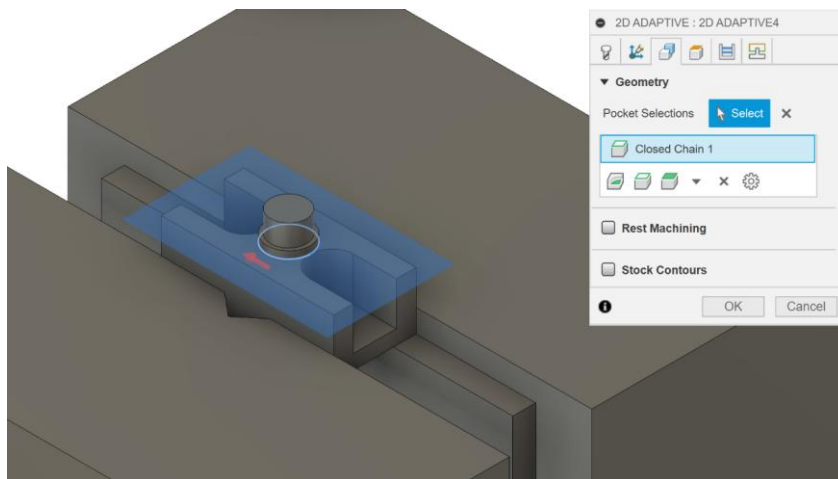


29. Double click on the setup to change the name to **OP20**.

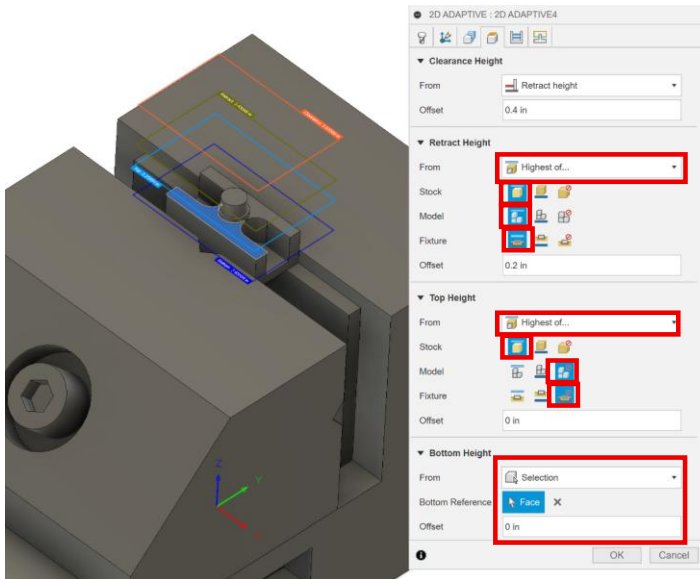
30. Select *2D-Adaptive Clearing*. Select *1/8" tool*.



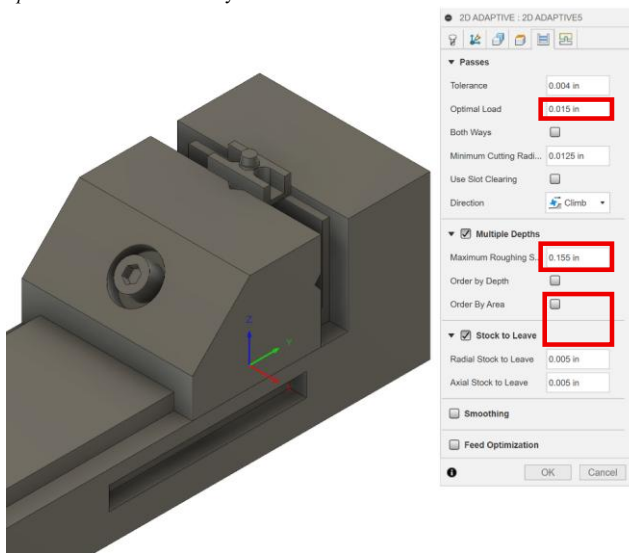
31. Under *geometry*, select the contour of the base of the cylinder that appears highlighted in the image.



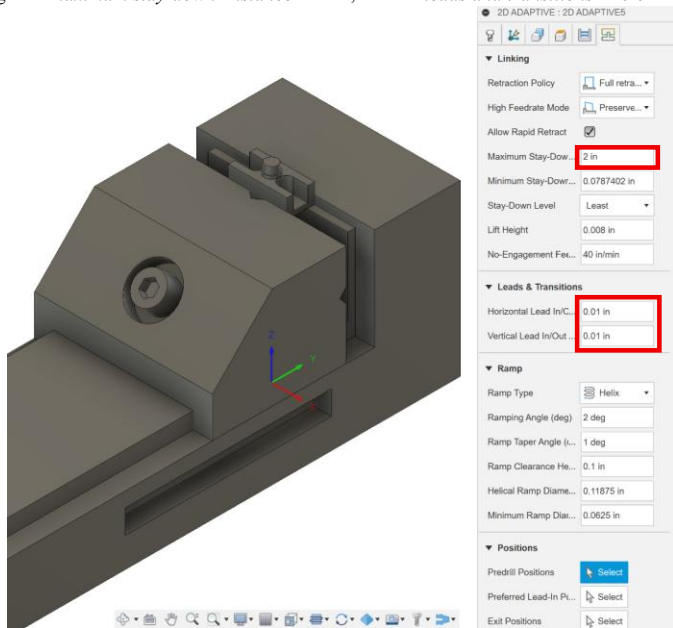
32. Change the *Retract Height* and *Top Height* to *highest of...* Change *bottom height* to *selection* and choose the top left side of the slider



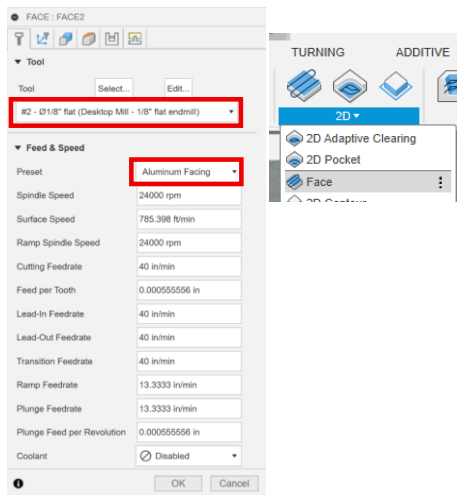
33. In the *passes* tab, change the *optimal load* to 0.015 in. Check the *multiple depths* and change the *maximum roughing stepdown* to 0.155 in. Modify the *radial stock* and *axial stock* to 0.005 in.



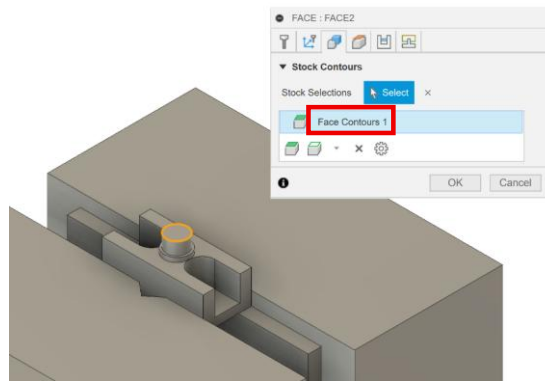
34. Change the *maximum stay-down Distance* to 2 in, and the *leads and transitions* to 0.01in. Then, click *ok*.



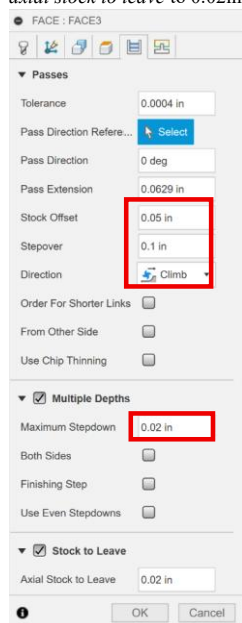
35. Select *2D- Face*. Select the *1/8" tool*. Then select *Aluminum facing*.



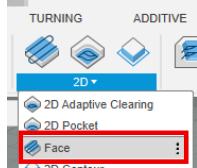
36. In the *geometry* tab, select the contour of the cylinder as shown in the picture.



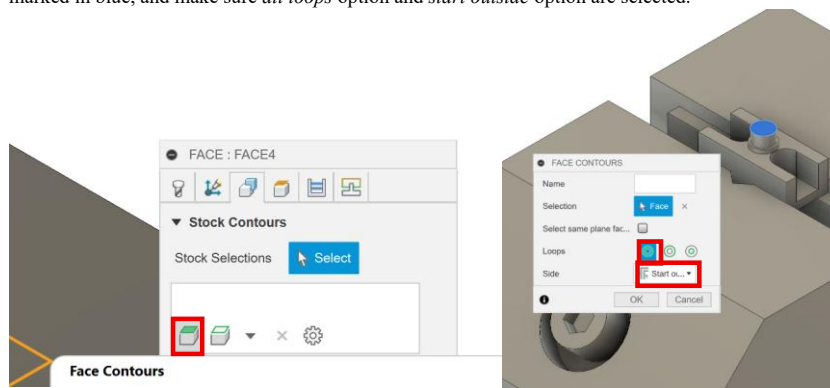
37. In the *passes* tab, change the *stock offset* to 0.05in, the *stepover* to 0.1 in and *direction* to climb. Then check *multiple depths* and change the *maximum stepdown* to 0.02in. Finally check the *stock to leave* and change the *axial stock to leave* to 0.02in. Click *ok*.



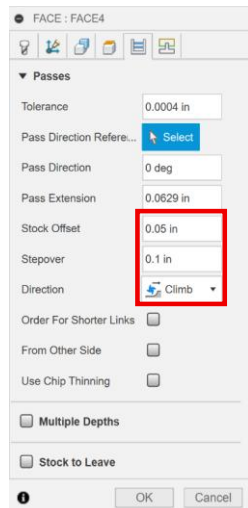
38. Create another *Facing* step. Keep the *Aluminum facing* preset.



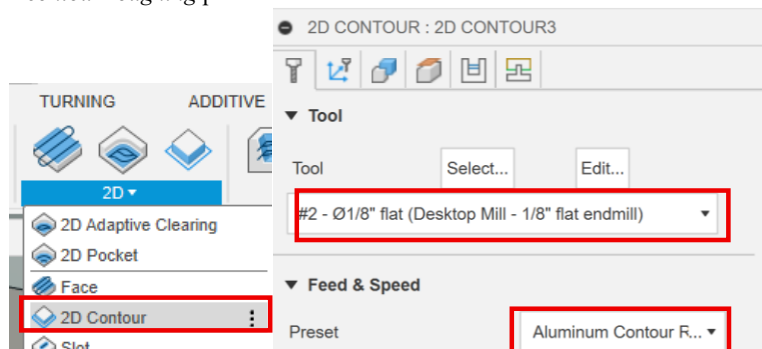
39. Click in the *geometry* tab and select *face contours*. A pop-up menu appears, select the top face of the cylinder marked in blue, and make sure *all loops* option and *start outside* option are selected.



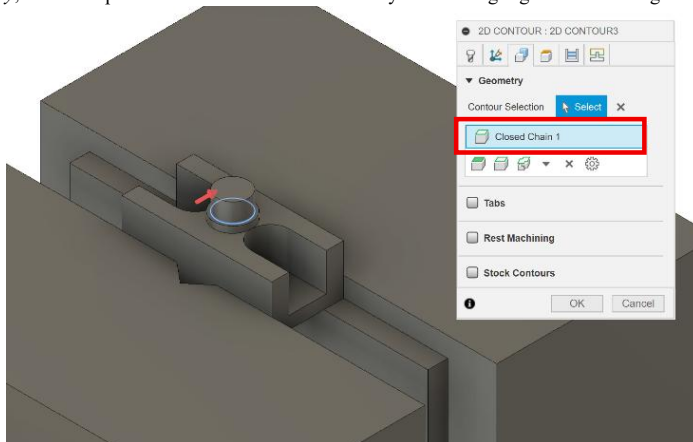
40. In the *passes* tab, change the stock offset to 0.05in, the stepover to 0.1 in and the direction to climb. Then, click ok.



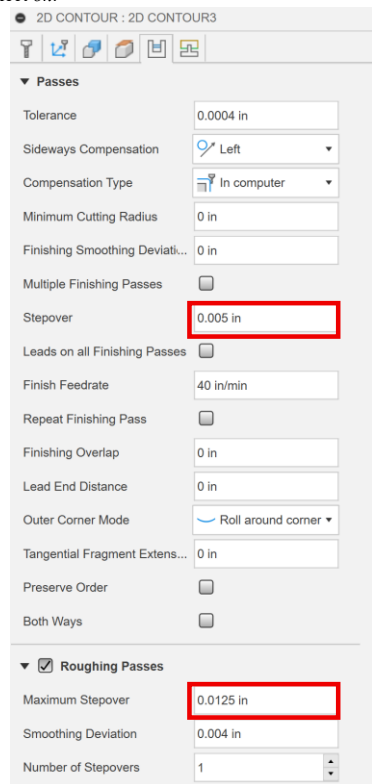
41. Add a 2D-Contour. Double check we are using the same tool. Choose the *Aluminum contour roughing* preset.



42. In *geometry*, select the perimeter of the base in the small cylinder as highlighted in the image.

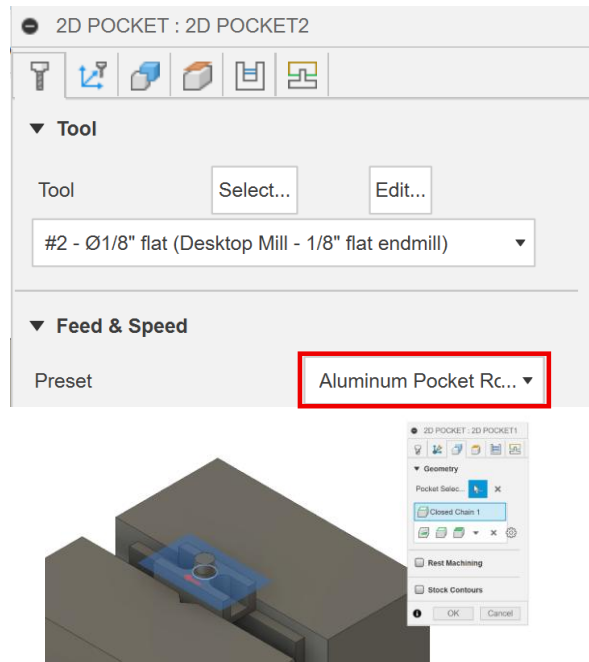


43. In the *passes* tab, check *roughing passes*. Change the *stepover* to 0.005in. Then, change the *maximum stepover* to 0.0125 in. Then select *ok*.

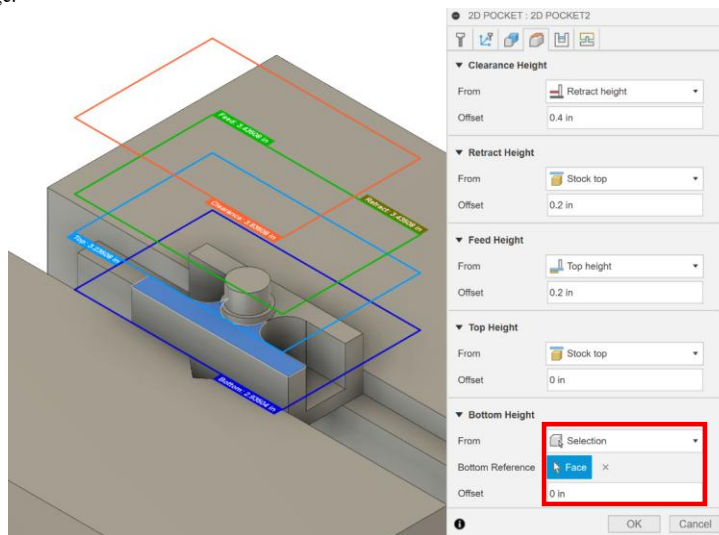


44. Add a *2D- Pocket*. Select the *Aluminum pocket roughing* preset. In the *geometry* tab, select the perimeter of the outer cylinder as highlighted in blue in the image.

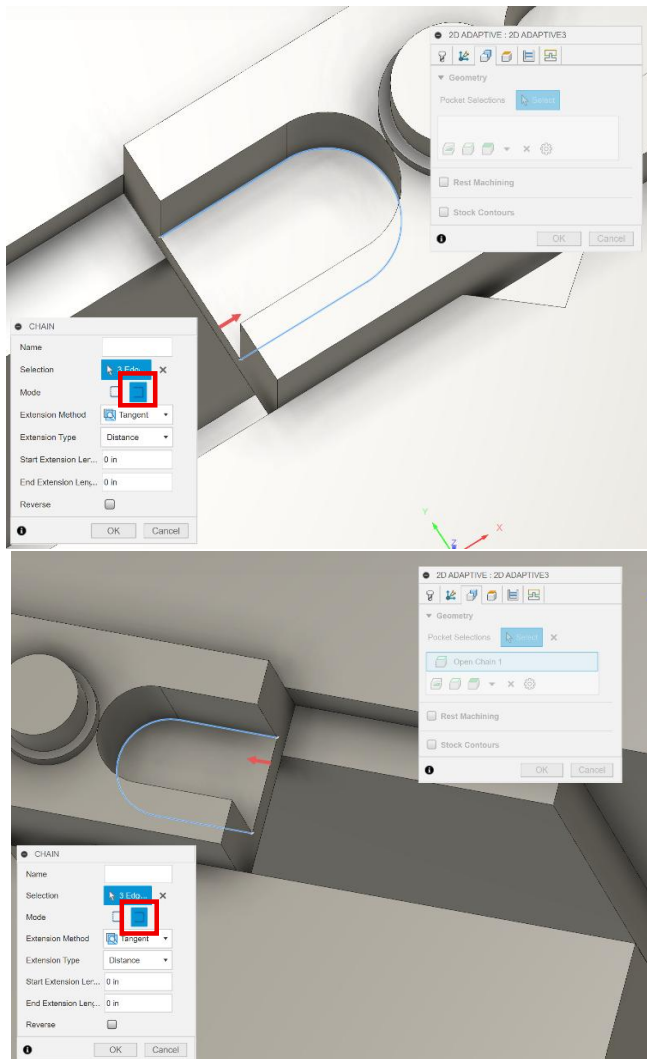




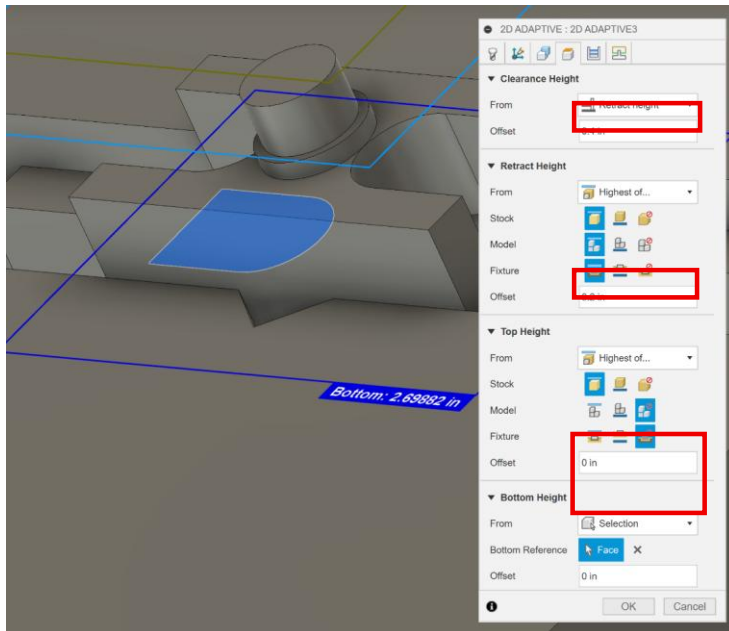
45. Under *heights* tab, *bottom height*, change from *option* to *selection* and select the top left surface as shown in the image.



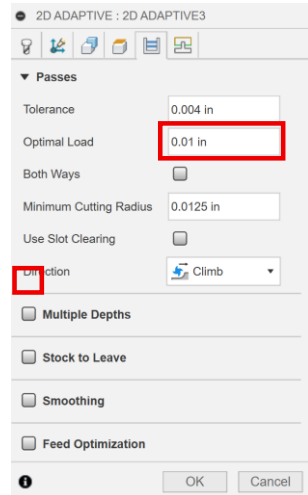
46. In *passes*, uncheck *stock to leave* option. Then click *ok*.



48. In *heights*, change the *retract height* and *top height* to *highest of*. For *bottom height*, switch from *option* to *selection* and select the highlighted bottom reference as shown in image. Orbit and zoom as needed.



49. In *passes*, change the *optimal load* to 0.01in and uncheck the *stock to leave* option.



50. In *linking*, change the *maximum stay-down distance* to 2 in and the *leads and transitions* to 0.01 in. Then click *ok*.

2D ADAPTIVE : 2D ADAPTIVE3

Linking

Retraction Policy: Full retraction

High Feedrate Mode: Preserve rapid movement

Allow Rapid Retract: ☒

Maximum Stay-Down Distance: 2 in

Minimum Stay-Down Clearance: 0.0787402 in

Stay-Down Level: Least

Lift Height: 0.008 in

No-Engagement Feedrate: 40 in/min

Leads & Transitions

Horizontal Lead In/Out Radius: 0.01 in

Vertical Lead In/Out Radius: 0.01 in

Ramp

Ramp Type: Helix

Ramping Angle (deg): 2 deg

Ramp Taper Angle (deg): 1 deg

Ramp Clearance Height: 0.1 in

Helical Ramp Diameter: 0.11875 in

Minimum Ramp Diameter: 0.0625 in

Positions

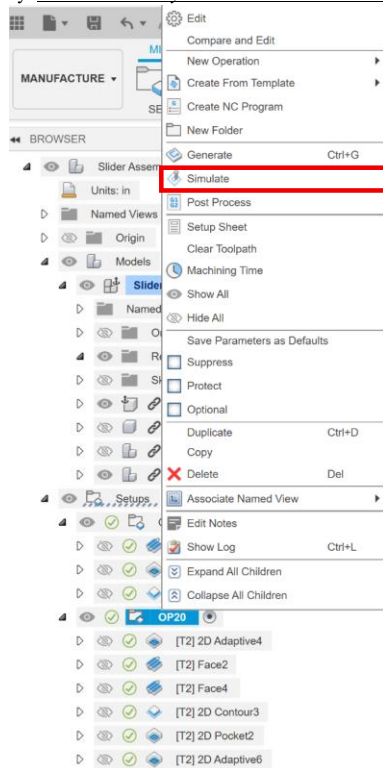
Predrill Positions: Select

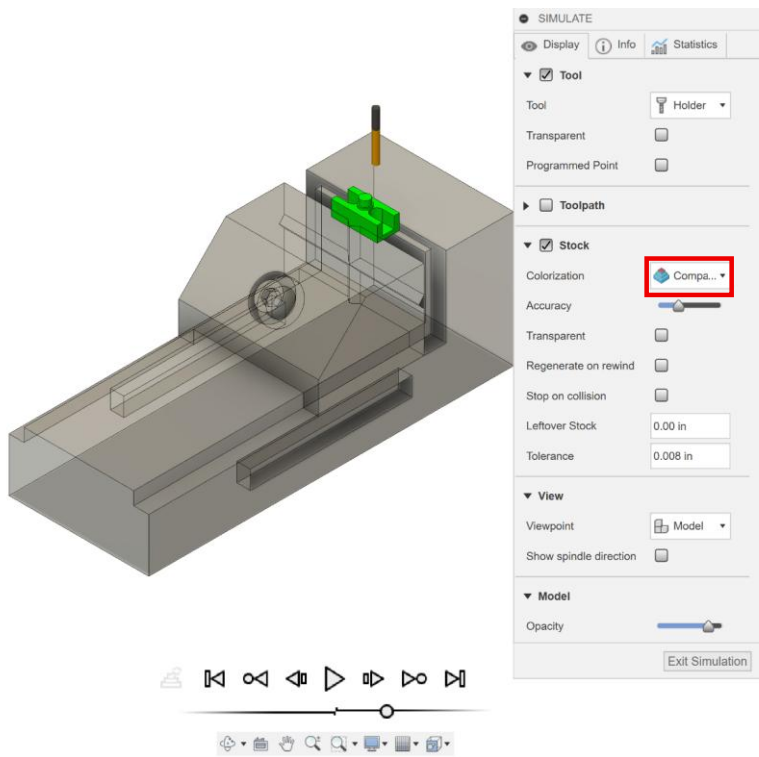
Preferred Lead-In Positions: Select

Exit Positions: Select

OK Cancel

51. *OP20* should now be finished. Hide *Sketches* in *Base Assembly v1:1*. Right click on *OP20* and select *Simulate*. Under *Colorization* choose *Comparison*. Press the play button at the bottom center of the window. Adjust the *Accuracy* slider until simulation runs smoothly. Once the simulation is complete, the part should be green if programmed correctly. Note: If *Accuracy* is set to the lowest value, blue may appear around the part.



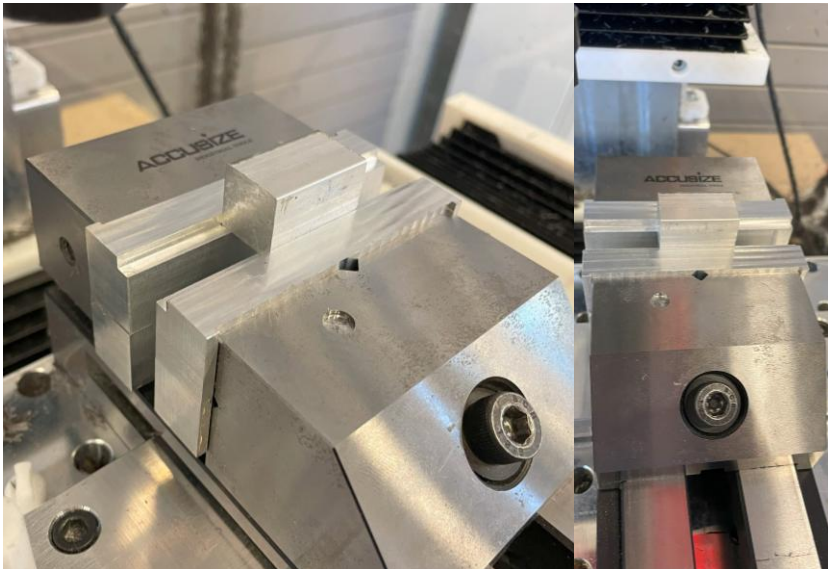


IV: Setup and Probing

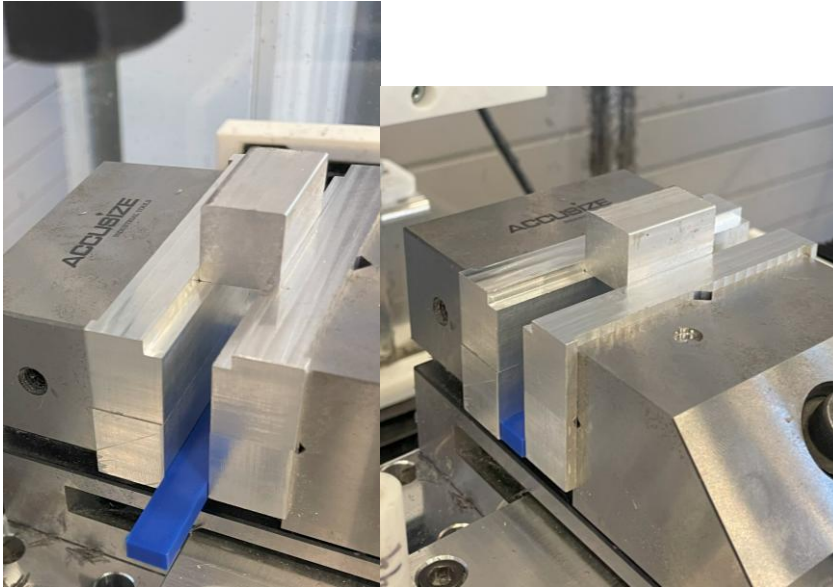
1. Place one aluminum soft jaw on each vise face, with the “ridge” side facing up and towards the other jaw (see below).



2. Place the slider stock centered on the inner ridge of the aluminum soft jaws. The stock should be resting with the widest face down on the parallels and the longest edge along the X-axis as seen below. Close the jaws but do not tighten.

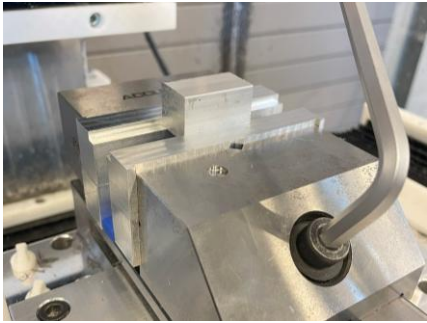


3. To prevent the jaws from moving when clamping a part (and because this vise has no threaded holes), place an appropriate plastic spacer between the soft jaw faces at the bottom of the jaws. The spacer should be about the same width as the gap in between the jaw faces, but loose enough to slide in. **This spacer is?? FIX FOR NEW SJ**



4. Use a 6mm Allen key to tighten the bolt in the center of the front vise jaw (if it is a different vise style, follow manufacturer instructions to clamp the part). This should close on the part and clamp tightly. As you tighten the vise, gently apply force down on the part. This will make sure that the part is evenly seated on the jaws (NOTE: an uneven surface may not seat perfectly on a flatter surface).

Check by hand to see if the part and spacer are tightly held. If the part is loose and the spacer is tight, choose a spacer of slightly smaller width. If the part is tight but the jaws have moved from flat against the jaw face, choose a slightly wider spacer. DO NOT attempt to run parts without good spacing and flat jaws: this could damage parts, tools, and the machine.



5. Gather the probe and attach the probing tip. Slide the probe into the spindle with a $\frac{1}{4}$ " collet and tighten by hand. This should hold the probe safely. Be sure that the spindle is turned off by the emergency stop or door switch.



6. With the electrical box on and connected to both the computer and machine tool, plug the probe into the electrical box and GENTLY move the probe tip. The light should change from green to red and the computer software should show a "P" alarm while the probe tip senses contact. Both conditions must be met to ensure probe operation.

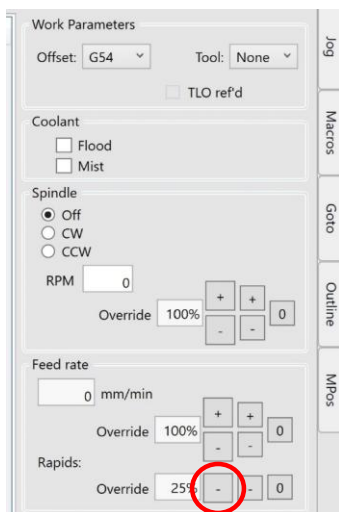


7. Carefully jog the probe over the part. Do not allow the probe tip to touch any surface while jogging, as excessive force or deflection can permanently damage the probe tip. Carefully jog the probe until it is as centered as visually possible over the part and 5-7 mm from the top surface of the part.

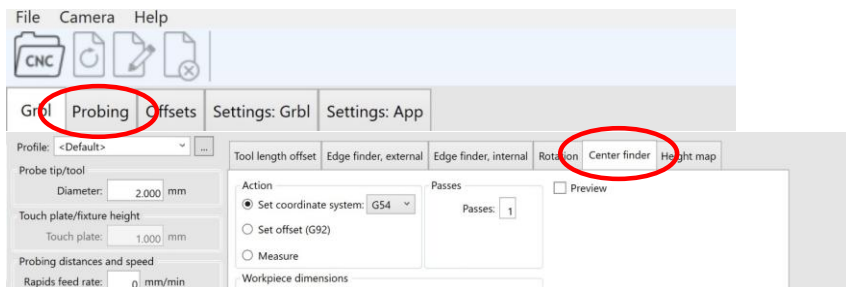


8. Before probing, reduce the rapids rate to 25%. Rapid movement rate can be changed with the minus sign buttons in the "Rapids" section shown below.

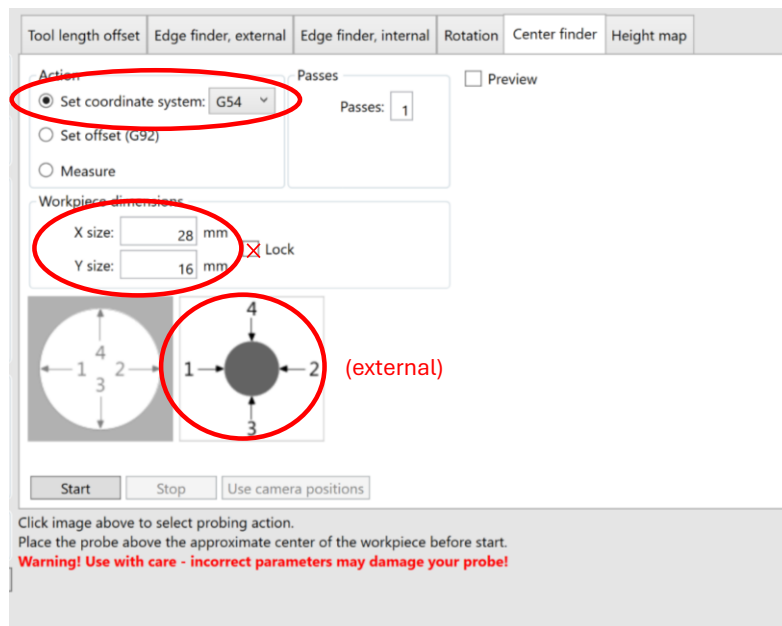
NOTE: this is an important step. This is done so that any errors in probing can be safely caught before any damage is done.



9. Move to the “Probing” tab in the ioSender software. From there, move to the “Center finder” option.



10. Ensure that the “Action” is set to “Set coordinate system” in G54. Uncheck the “Lock” box. Measure the stock and type in the workpiece dimensions in millimeters. This should be roughly 28mm in the X-direction and 16mm in the Y-direction. Select the external option. Select the external bore option for probing the outside surfaces of the stock.



- In the "Probing clearances" section, set the probing depth. This value will determine how far down the probe travels to touch the workpiece. It is equal to the distance between the current probe tip position and the place on the part that will be probed-- it may be necessary to roughly measure this distance before probing to ensure correct positioning and mitigate errors.

Probing clearances

XY Clearance: mm

Offset: mm

Depth: mm

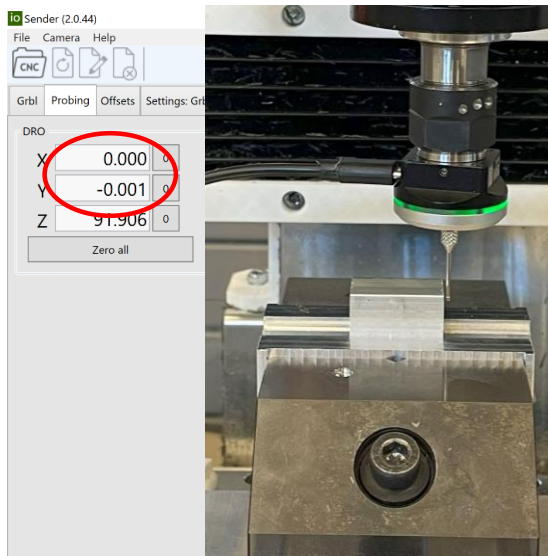
Probe XY offsets

X: mm Y: mm

Click to activate keyboard jogging

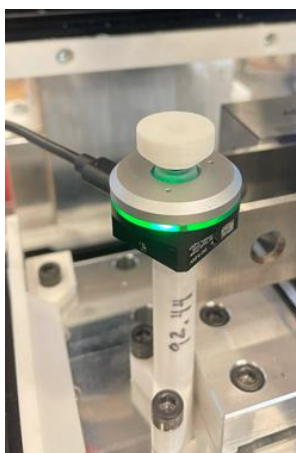
X:-21.497 Y:-6.674 Z:86.209

12. Double check inputs. Errors can be costly and are much more difficult to catch quickly in operation. Press “Start” and watch the probing cycle very closely. Emergency stop the machine if anything unexpected occurs and safely try again. The machine should move across the part to touch each side once. This will set the X and Y offsets for a coordinate system centered on the rectangular block. The X- and Y- offsets should be zeroed.



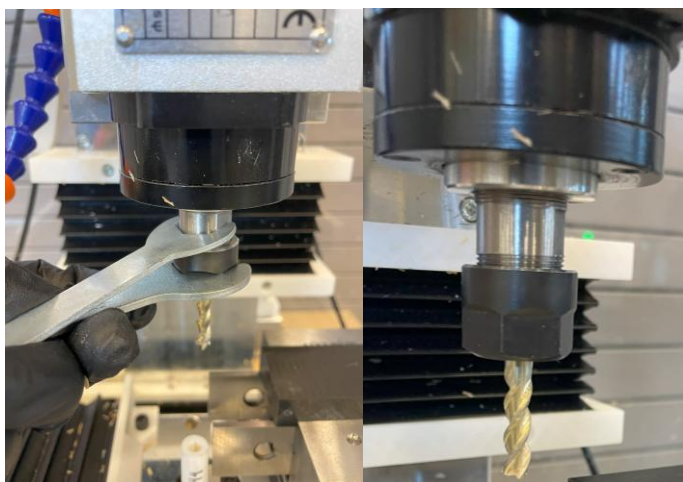
13. We will now set the Z height for the coordinate system. This is based on the surface of the X-carriage and will include the tool offset. Jog the probe safely away from the part and remove it from the spindle.

14. Gently replace the probe tip with the probe touch plate. Insert the shaft of the probe into the holder on the left side of the vise. If the signal cable was disconnected, plug it into the probe and test again to be sure the signal is working.



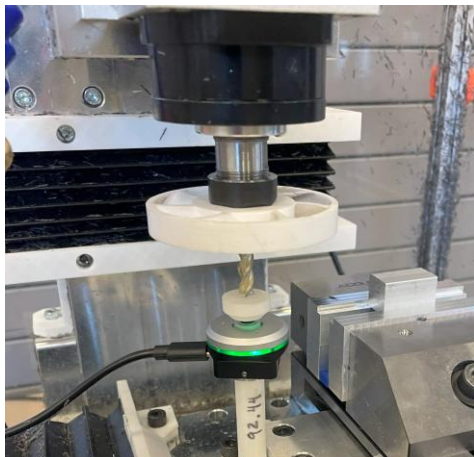
15. Press the emergency stop. **Never handle tools in the spindle without the emergency stop engaged.**

Insert the $\frac{1}{4}$ " endmill into the spindle and secure tightly using the provided wrenches. The tool should be held just above the flutes.



16. NOTE: Trying to jog any axis while the emergency stop is engaged will move the coordinate system without moving the machine. Do not jog until the emergency stop is disengaged.

Carefully jog the spindle until the tool rests approximately 5mm above and centered on the touch plate.



17. Move back to the “Probing” tab and select “Tool length offset”. Check the “Add offset” box and set the touch plate height to 92.44 mm (or the otherwise determined accurate distance from the touch plate to the X-carriage surface – see *Troubleshooting*).

Profile: <Default> ...

Probe tip/tool

Diameter: 2.000 mm

Touch plate offset height

Touch plate: 92.44 mm

Probing distances and speed

Rapids feed rate: 0 mm/min

Search feed rate: 100 mm/min

Latch feed rate: 25 mm/min

Probing distance: 10.000 mm

Latch distance: 0.500 mm

Probing clearances

XY Clearance: 5.000 mm

Offset: 5.000 mm

Depth: 9 mm

Probe XY offsets

X: 0.000 mm Y: 0.000 mm

Tool length offset | Edge finder, external | Edge finder, internal | Rotation | Center finder | Height map

☐ Probe fixture @ G59.3 with tool: ...

☒ Establish reference offset

Workpiece offset

☒ Add offset

Action

☒ Set coordinate system: G54

Workpiece Height: 0.000 mm

☐ Set offset (G92)

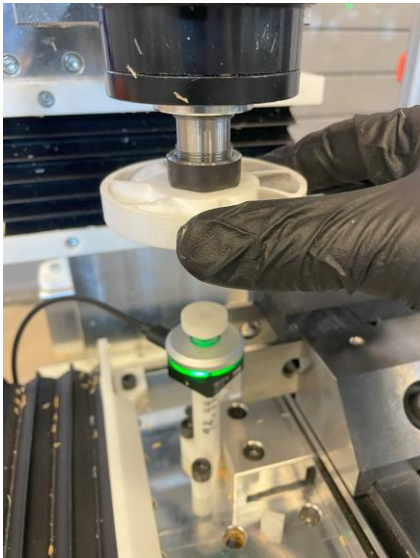
Start Stop

Warning! Use with care - incorrect parameters may damage your probe!

18. Double check all parameters and press start. The tool should slowly approach the touch plate and gently press it twice. This will set the Z-coordinate at the bottom of the vise / top of the X-carriage.



19. Jog the tool out of the way and remove the probe completely from the enclosure. If the machine tool is not equipped with compressed air, slide the plastic chip fan onto the spindle over the tool as seen below.



20. Remove all tools or obstacles from the enclosure and replace the door.

21. Jog the tool so that it starts the operation **above** the part.

The machine tool should now be ready to run the first operation.

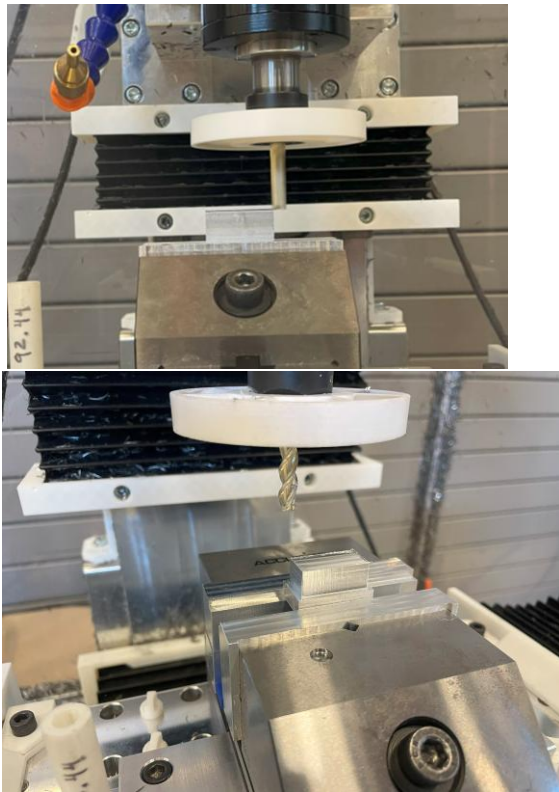
REVIEW AND FOLLOW GENERAL INSTRUCTIONS AND CHECKLIST.

Be ready to feed hold the machine IMMEDIATELY after pressing cycle start. This will allow the spindle to reach the desired speed before milling begins (this usually takes no longer than 5 seconds – the spindle should be making a constant sound and not “ramping up”).

ALWAYS ALLOW THE SPINDLE TO GET UP TO SPEED BEFORE ENTERING A CUT.

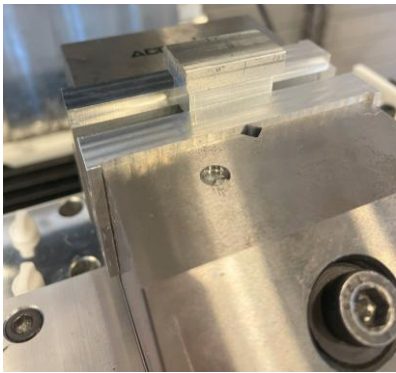
After pressing feed hold and letting the spindle reach its target RPM, you can resume the operation with cycle start.

22. Watch the entire operation carefully.

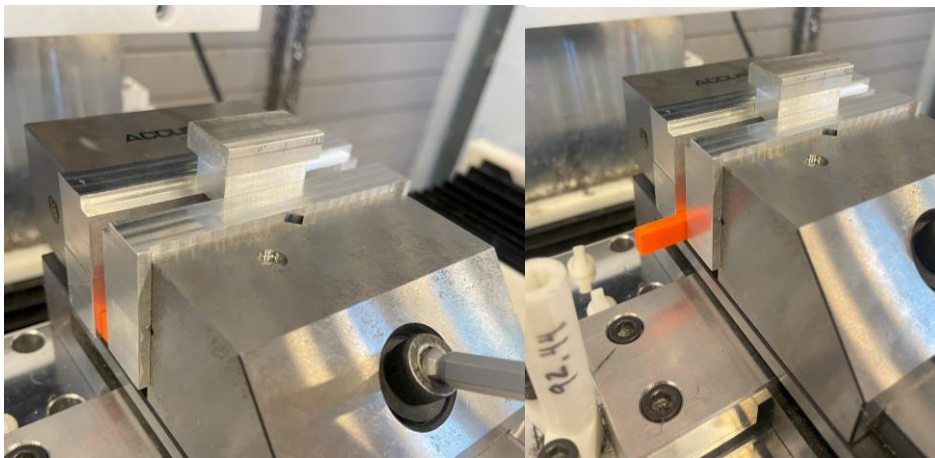


OP20

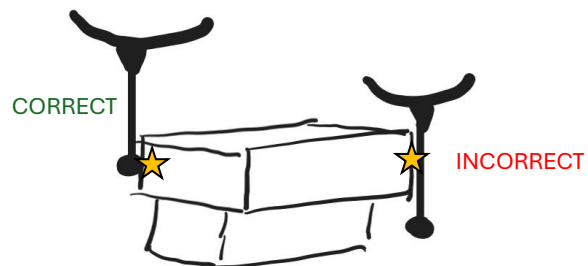
23. Once OP10 is complete, jog the tool out of the way. Emergency stop the machine and open the door. Clear any chips as necessary. Carefully slide the chip fan off the spindle. Remove the 1/4" endmill with the wrenches.
24. Use the Allen key to loosen the vise jaw and remove the cut part.
25. Replace the part upside down, on the same soft jaws. You will clamp down on the newly cut surface in the same orientation as before.



26. Repeat steps 1-4 with a new spacer. In this example, a 22mm spacer was used.



27. Follow steps 5-21 to probe the part. The X- and Y- measurements should remain the same. For this setup, ensure that the probing depth is set so that the flat faces of the stock contact the probe tip and not the probe shaft while probing.

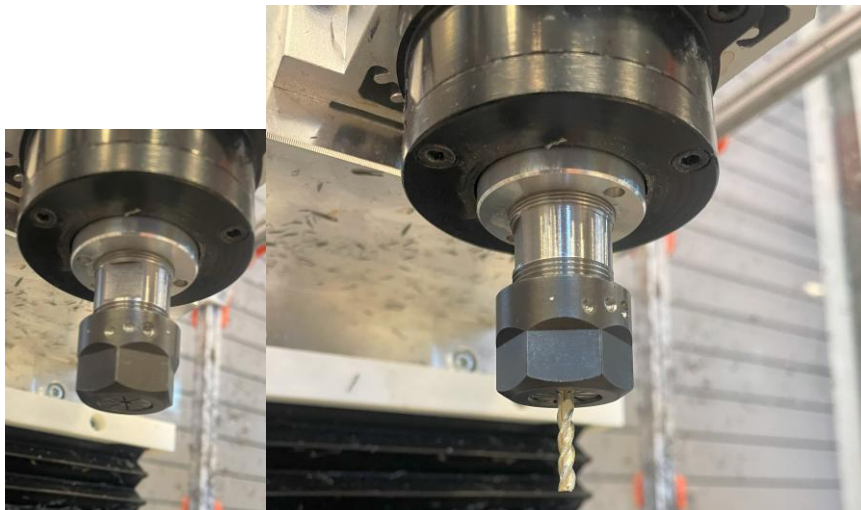


28. Press the emergency stop. **Never handle tools in the spindle without the emergency stop engaged.**

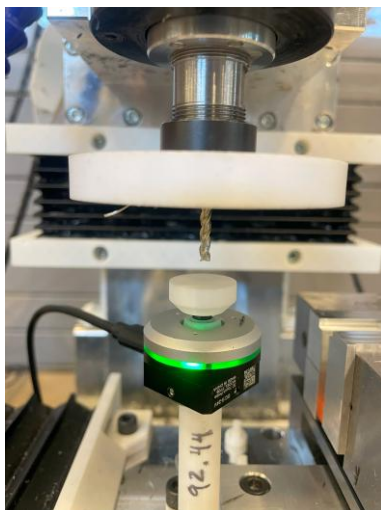
Fully unscrew the collet from the spindle. Gently press out the $\frac{1}{4}$ " collet and replace with a $\frac{1}{8}$ " collet. Press until the collet "clicks" into place and is correctly seated.



29. Screw the collet onto the spindle loosely, then insert the 1/8" endmill into the spindle and secure tightly using the provided wrenches. The tool should protrude just past the start of the flutes.

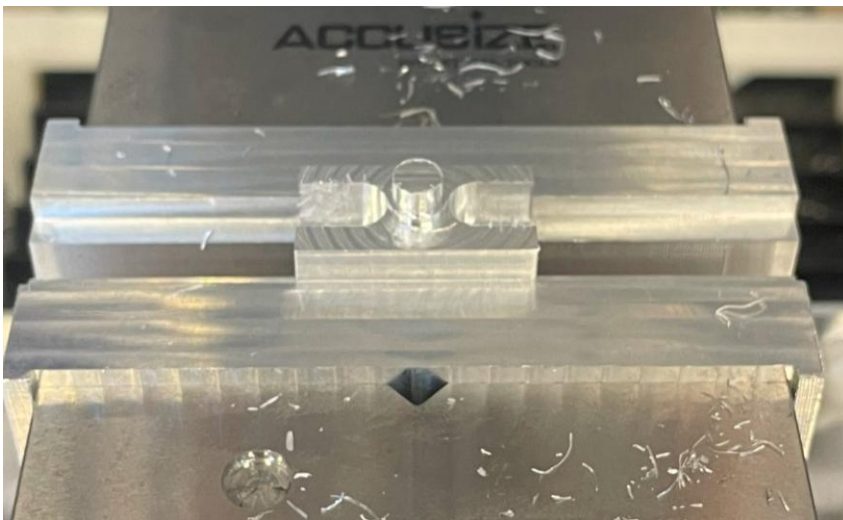
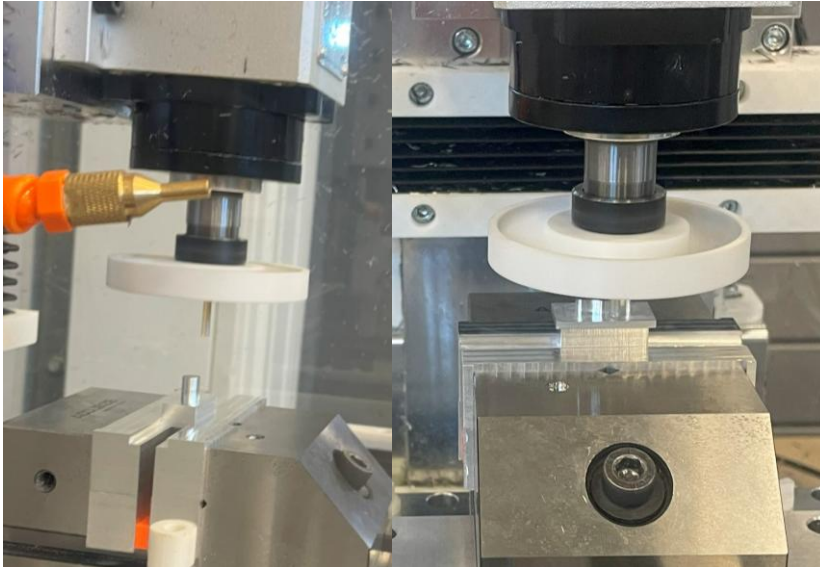


30. Repeat steps X-X with the new tool.



31. Repeat step X.

32. Watch the entire operation carefully.



33. Once OP20 is complete, jog the tool out of the way. Emergency stop the machine and open the door. Clear any chips as necessary. Carefully slide the chip fan off the spindle. Remove the 1/8" endmill with the wrenches.
34. Use the Allen key to loosen the vise jaw and remove the cut part.